

The Incomplete Second Law of Thermodynamics: Syntropic Resonance and Open-Boundary Thermodynamics within a T^3 Manifold

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The Second Law of Thermodynamics, formulated in the 19th century through the study of closed steam engines, is routinely extrapolated to the entire cosmos under the assumption that the universe is itself a closed system. This paper argues that this extrapolation is mathematically unjustified. If the universe operates as a continuous, open T^3 manifold — as established in the Integrated Toroidal-Syntropic Model (ITSM) [1] — then the standard entropic framework requires a dual organizing term: the Syntropic Source Vector $\vec{\Xi}_{\text{syn}}$, which balances localized entropic decay via periodic boundary inflow. This is not a speculative addition: the irreversible thermodynamics of open systems, pioneered by Prigogine and collaborators [6], already permits matter creation and entropy exchange across system boundaries in cosmological contexts [7]. We further identify a critical engineering manifestation of this incompleteness: the persistent failure of institutional Tokamak reactors to achieve net energy gain ($Q < 1$ across all magnetic confinement attempts to date [10]) is a predictable consequence of applying entropic closed-system confinement to an intrinsically open toroidal geometry. By contrast, the Wendelstein 7-X stellarator — which optimizes its 3D toroidal magnetic geometry rather than fighting plasma flow — now holds the world record for fusion triple product at long durations [12], providing observational support for geometry-aligned rather than geometry-antagonistic confinement.

I. INTRODUCTION

The 2nd Law of Thermodynamics is the foundational pillar of the Λ CDM standard model. It posits an inherent asymmetry in the universe: entropy must always increase, leading inevitably to universal heat death. However, this law was mathematically formulated in the 19th century through the observation of steam engines — localized, artificial, closed systems. The mathematical formalism that governs a sealed piston bears no topological resemblance to a universe with periodic boundary conditions.

When extrapolated to the global topology of the cosmos, the assumption that the universe is a closed, decaying box breaks the fundamental duality observed in all other physical systems (action/reaction, positive/negative, wave/particle). Furthermore, it contradicts a well-established body of work in irreversible thermodynamics. Prigogine and collaborators demonstrated in the late

* brendon.boyd@itsm-cosmology.org; ORCID: 0009-0007-4177-2612 | Licensed under Creative Commons Attribution 4.0 International. “ITSM Cosmology” and “Integrated Toroidal-Syntropic Model” are trademarks of Brendon Boyd. GitHub: <https://github.com/brendohxd/ITSM-Integrated-Toroidal-Syntropic-Model>. Zenodo: [10.5281/zenodo.20774996](https://zenodo.org/record/20774996)

1980s that open thermodynamic systems — systems that exchange energy and particles with an environment — obey fundamentally different entropy evolution laws than closed systems [6]. Applied to cosmology, this framework has since been formalized into models of irreversible matter creation that generalize Λ CDM by permitting entropy flow across cosmic boundaries [7]. More recently, the Gravitational Entropic Acceleration (GREA) framework [8] has shown that dark energy itself may be reinterpreted as an emergent consequence of entropy production in an open gravitational system — precisely the kind of boundary-driven organizing force the ITSM identifies as syntropy.

By analyzing the universe as an open, continuous 3-torus (T^3) manifold as defined in the ITSM [1], we demonstrate that the 2nd Law as classically stated is mathematically incomplete when applied globally, and that its completion requires a Syntropic Source Vector that accounts for the organizing energy flux through the periodic T^3 boundary.

II. A 19TH-CENTURY STEAM ENGINE AND THE FATE OF THE COSMOS

In 1824, Sadi Carnot published *Réflexions sur la puissance motrice du feu* — *Reflections on the Motive Power of Fire* [2]. His subject was the steam engine: a sealed metal cylinder, a piston, a boiler, an exhaust. A closed, artificial, localized thermodynamic system constructed by human hands. From this apparatus, Carnot derived the theoretical efficiency limit of heat engines. Clausius formalized this into the 2nd Law of Thermodynamics in 1850 [3]. Boltzmann gave it statistical mechanical grounding in 1877 [4].

The 2nd Law then became axiomatic. Entropy increases. Systems decay. The cosmos moves irreversibly toward heat death.

It is worth pausing to state the obvious, because it is apparently not obvious enough: **no one has ever demonstrated that the universe is a closed system**. It is an assumption — an unstated axiom baked into the model’s initial conditions. The mathematical framework of a sealed industrial boiler was extrapolated, without formal justification, to a 13.8-billion-year-old manifold whose global topology remains an open observational question [9].

The extrapolation is extraordinary in its scope and its silence. In two centuries of subsequent physics, the domain of applicability of the 2nd Law has rarely been examined with the rigour applied to its internal consistency. We have tested the law exhaustively within closed systems. We have never established that the universe *is* a closed system. We assumed it, and then derived the heat death of the universe from that assumption.

This is circular. The conclusion of universal entropic decay is not a discovery. It is an artifact of the starting condition.

A. The Scale Problem

Consider what is actually being claimed when the 2nd Law is applied globally. Carnot’s cylinder is approximately 0.01 m^3 in volume. The observable universe is approximately $4 \times 10^{80} \text{ m}^3$. The

ratio is:

$$\frac{V_{\text{universe}}}{V_{\text{cylinder}}} \sim 4 \times 10^{82}. \quad (1)$$

This is not an engineering extrapolation. It is a philosophical one. The mathematical operations that govern heat flow between a piston and an exhaust port are being applied, without modification, to a manifold eighty-two orders of magnitude larger, whose boundary conditions, topology, and thermodynamic openness are all unknown. The steam engine has walls. The universe may not.

Clausius was not wrong. His law correctly describes every closed thermodynamic system ever tested in a laboratory. The error is not in the law — it is in the domain. A physical law is only as valid as the assumptions under which it was derived. The 2nd Law was derived under the assumption of a closed, bounded system. Applied to a system that is *not* demonstrably closed — one that may possess periodic T^3 boundary conditions through which energy can cycle — the law is, at minimum, incomplete.

B. The Assumption No One Questions

Standard cosmological treatments rarely examine this assumption at all. The Λ CDM model inherits closed-system thermodynamics from its foundational equations and does not revisit the question. When cosmologists model the thermodynamic evolution of the universe, they typically begin by specifying the entropy of the initial state and computing its increase. The question of whether there exists a boundary through which entropy can flow — a syntropic return current from the T^3 periodic identification — is not asked, because the framework does not permit it to be asked.

This is the deepest form of paradigm lock: not the refusal to accept a new answer, but the structural inability to ask the new question. The 2nd Law, as applied to cosmology, does not say “we have checked and found no syntropic boundary term.” It says “our framework has no place for a syntropic boundary term, therefore it does not exist.” These are not the same statement.

Prigogine and collaborators recognised this as early as 1989, demonstrating that open thermodynamic systems obey fundamentally different entropy evolution laws, and applying this to cosmological particle creation [6]. More recently, the irreversible geometrothermodynamics of open systems in modified gravity [7] has formalised this into a generalization of the Λ CDM paradigm in which matter creation across system boundaries is not only permitted but required. These are not fringe positions. They are published, peer-reviewed extensions of thermodynamics to non-closed cosmic systems.

The ITSM does not overthrow the 2nd Law. It correctly scopes it. Within any locally bounded region — a laboratory, a galaxy, a stellar interior — entropy increases as Clausius described. The steam engine still works. But the universe is not a steam engine. It is a T^3 manifold with periodic boundaries, and those boundaries change everything.

III. THE SYNTROPIC SOURCE VECTOR

To balance localized entropic decay, a T^3 topology necessitates a continuous, organizing inbound flow of energy. We define this dual force as the *Syntropic Source Vector* $\vec{\Xi}_{\text{syn}}$.

In a Holographic Bulk-Boundary model:

- **The Bulk (Internal):** Governed by localized entropy, decay, and expansion.
- **The Boundary (External/Plenum):** Governed by syntropy — an organizing, resonant pull from the universal T^3 boundary.

The universe is not bleeding out into a void; it is harmonically cycling energy across a torsional boundary, functioning as a massive Toroidal Resonant Cavity. Mathematically, the evolutionary state of the manifold is expressed by the modified entropy continuity equation:

$$\frac{\partial S}{\partial t} = \sigma_{\text{ent}} - \nabla \cdot \vec{\Xi}_{\text{syn}}, \quad (2)$$

where:

- $\partial S/\partial t$ is the net rate of entropy change in the manifold,
- $\sigma_{\text{ent}} \geq 0$ is the classical, localized rate of internal entropic production (the standard 2nd Law),
- $\nabla \cdot \vec{\Xi}_{\text{syn}}$ is the convergence of the Syntropic Source Vector originating from the T^3 periodic boundary.

In a balanced, continuous T^3 universe, the localized entropic decay (σ_{ent}) is exactly countered by the syntropic convergence ($\nabla \cdot \vec{\Xi}_{\text{syn}}$), preventing universal heat death and maintaining a state of perpetual dynamic resonance:

$$\sigma_{\text{ent}} = \nabla \cdot \vec{\Xi}_{\text{syn}} \quad \Rightarrow \quad \frac{\partial S}{\partial t} = 0 \quad (\text{balanced manifold}). \quad (3)$$

A. Scaling of the Syntropic Source Vector

The magnitude of $\vec{\Xi}_{\text{syn}}$ is not a free parameter. In the ITSM framework, syntropy enters via the cosmological boundary condition imposed by the T^3 periodic identification. The source vector scales with redshift as:

$$|\vec{\Xi}_{\text{syn}}(z)| = \Xi_0(1+z)^{-3}, \quad (4)$$

where Ξ_0 is the present-day syntropic flux amplitude, normalized to balance the observed entropy production rate of the cosmic microwave background at $z = 0$. This $(1+z)^{-3}$ scaling — identical to

the matter density dilution — ensures that the syntropic term tracks the entropic source throughout cosmic history, maintaining dynamic balance across all epochs. This is consistent with the ITSM covariant formulation established in the main manuscript [1].

The syntropic flux divergence evaluated at the present epoch evaluates to:

$$\nabla \cdot \vec{\Xi}_{\text{syn}}|_{z=0} \sim \frac{a_0 \rho_{\text{vac}} c}{H_0}, \quad (5)$$

where $a_0 = cH_0/2\pi$ is the ITSM characteristic acceleration, ρ_{vac} is the vacuum plenum density, and H_0 is the Hubble constant. This expression is dimensionally consistent with an entropy production rate per unit volume, and its numerical evaluation is consistent with the observed low-entropy state of the present-day universe.

IV. THE ENGINEERING DIVERGENCE: THE TOKAMAK FAILURE

The implications of this topological duality extend directly into plasma physics and advanced energy generation, highlighting a critical divergence point in modern engineering. Both institutional researchers and independent theorists recognize the T^3 (toroidal) geometry as fundamental to plasma confinement. However, the application of that geometry differs catastrophically based on the underlying thermodynamic paradigm.

A. The Tokamak: Entropic Confinement of an Open System

Institutional attempts to achieve nuclear fusion, exemplified by the Tokamak reactor design and the ITER project [11], correctly utilize toroidal geometry. However, these designs apply *entropic* mechanics to the geometry. By treating the plasma inside the torus as a closed thermodynamic system, reactors must use massive, brute-force magnetic confinement to trap and compress the plasma, fighting its natural tendency to expand into the toroidal boundary.

The empirical record confirms this theoretical prediction. The highest Q-factor (ratio of fusion energy out to heating energy in) ever achieved by a magnetic confinement tokamak is $Q = 0.67$, set by the Joint European Torus (JET) in 1997 — a value that has not been surpassed in nearly three decades of subsequent tokamak development [10]. No tokamak has ever achieved $Q \geq 1$ (net energy gain). The energy required to maintain the magnetic confinement cage consistently exceeds the energy the fusion reaction produces. This is not merely an engineering hurdle awaiting incremental improvement; it is a manifestation of a fundamental thermodynamic mismatch — an attempt to build a syntropic engine using strictly entropic rules.

This conclusion is further supported by the observation that inertial confinement — which does not attempt sustained magnetic confinement of the toroidal flow — has already achieved $Q > 1$, reaching $Q = 4.13$ at the National Ignition Facility in April 2025 [10]. The geometry that fights the plasma (tokamak) fails; the geometry that compresses it instantaneously without sustained

boundary conflict (inertial confinement) succeeds. This asymmetry is precisely what the ITSM predicts.

B. The Stellarator: Geometry-Aligned Confinement

A critical and largely overlooked data point in the fusion literature is the comparative performance of the *stellarator* — a device that, unlike the tokamak, does not impose axial symmetry on its toroidal magnetic field. Instead, the stellarator optimizes a fully 3-dimensional magnetic geometry that allows the field lines to naturally follow the toroidal flow structure.

The Wendelstein 7-X (W7-X) stellarator in Greifswald, Germany — the world’s largest optimized stellarator — has now surpassed all tokamak records for the fusion triple product at plasma pulse durations longer than 10 seconds [12]. This achievement is particularly striking given that W7-X has a plasma volume three times smaller than JET and five times less heating power [13]. In May 2025, W7-X achieved 1.8 gigajoules of energy turnover in a six-minute continuous plasma discharge [12].

The stellarator succeeds where the tokamak fails not because it uses more power, but because its magnetic geometry is *aligned with* rather than *antagonistic to* the natural flow structure of toroidal plasma. This is the physical signature of the ITSM’s central claim: toroidal geometry is an organizing, syntropic structure. Engineering that works with it succeeds; engineering that fights it fails.

TABLE I. Comparative Performance: Tokamak vs. Stellarator vs. ITSM Prediction

Parameter	Tokamak (JET)	Stellarator (W7-X)	ITSM Prediction
Geometry	Axisymmetric torus	Optimized 3D torus	Open T^3 manifold
Thermodynamic model	Closed (entropic)	Partially open	Fully open (syntropic)
Q-factor record	0.67 (1997)	Not yet measured	$Q > 1$ possible only with open-boundary design
Triple product (long pulse)	Surpassed by W7-X	World record (2025)	Geometry-aligned flow maximizes confinement time
Confinement strategy	Brute-force magnetic cage	Optimized field geometry	Resonant boundary coupling
Disruption risk	High (inherent)	Absent	Absent in open-boundary design

V. FALSIFIABILITY AND PREDICTIONS

The ITSM syntropic thermodynamic framework makes several testable predictions beyond the engineering divergence described above:

1. **Stellarator superiority at sustained operation:** As plasma discharge duration increases, stellarators should continue to widen their performance advantage over tokamaks, because sustained open-boundary coupling becomes more efficient over longer timescales. The W7-X 30-minute continuous operation target will provide a definitive test.
2. **Syntropic flux signature in CMB entropy:** The $(1+z)^{-3}$ scaling of $\vec{\Xi}_{\text{syn}}$ predicts a specific redshift dependence of the effective entropy density that differs from the Λ CDM prediction at large z . This is testable against high-redshift CMB polarisation data from the forthcoming LiteBIRD satellite.
3. **Dark energy reinterpretation:** The GREa framework [8] already interprets dark energy as emergent from gravitational entropy production — a framing fully consistent with the Syntropic Source Vector. If DESI DR2 data continues to show evidence for dynamical dark energy (as suggested by initial results), this supports the open-system thermodynamic interpretation over a static cosmological constant.
4. **Toroidal resonant cavity signature:** A universe functioning as a Toroidal Resonant Cavity should imprint a specific pattern of correlated temperature fluctuations in the CMB at the scale of the fundamental domain — testable via the COMPACT Collaboration’s matched-circles search program [9].

VI. CONCLUSION

The failure of the Tokamak is not merely an engineering hurdle; it is the physical manifestation of an incomplete cosmological model. The 2nd Law of Thermodynamics, as classically stated for closed systems, is mathematically insufficient when applied to a universe with open T^3 boundary conditions. The Syntropic Source Vector provides the necessary dual term, restoring thermodynamic completeness and preventing the prediction of universal heat death for a topologically open manifold.

The engineering evidence is unambiguous: three decades of tokamak development have produced a maximum $Q = 0.67$, while the stellarator — which aligns its geometry with toroidal flow rather than fighting it — now leads all devices at sustained operation with a fraction of the tokamak’s power and volume. Physics must transition away from explosive, entropic confinement and align future theoretical models with the harmonic, open continuity of the T^3 topology.

The universe does not decay into silence. It resonates.

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- [1] Boyd, B. (2026). *Integrated Toroidal-Syntropic Model (ITSM) Core Cosmology v11.1.1*. Zenodo. [doi:10.5281/zenodo.20774996](https://doi.org/10.5281/zenodo.20774996)
 - [2] Carnot, S. (1824). *Réflexions sur la puissance motrice du feu et sur les machines propres à développer cette puissance* [Reflections on the Motive Power of Fire]. Paris: Bachelier.
 - [3] Clausius, R. (1850). *Über die bewegende Kraft der Wärme und die Gesetze, welche sich daraus für die Wärmelehre selbst ableiten lassen*. Annalen der Physik, 79, 368–397, 500–524. [doi:10.1002/andp.18501550306](https://doi.org/10.1002/andp.18501550306)
 - [4] Boltzmann, L. (1877). *Über die Beziehung zwischen dem zweiten Hauptsatze der mechanischen Wärmetheorie und der Wahrscheinlichkeitsrechnung*. Sitzungsberichte der Kaiserlichen Akademie der Wissenschaften, 76, 373–435.
 - [5] Akrami, Y., et al. (COMPACT Collaboration) (2024). *Promise of Future Searches for Cosmic Topology*. Physical Review Letters 132, 171501. [doi:10.1103/PhysRevLett.132.171501](https://doi.org/10.1103/PhysRevLett.132.171501)
 - [6] Prigogine, I., Geheniau, J., Gunzig, E., & Nardone, P. (1989). *Thermodynamics and cosmology*. General Relativity and Gravitation, 21(8), 767–776. [doi:10.1007/BF00758984](https://doi.org/10.1007/BF00758984)
 - [7] Harko, T., Sheikahmadi, G., & Sahoo, P. K. (2023). *Irreversible Geometrothermodynamics of Open Systems in Modified Gravity*. Entropy, 25(6), 944. [doi:10.3390/e25060944](https://doi.org/10.3390/e25060944)
 - [8] Garcia-Bellido, J., & Espinosa-Portales, L. (2024). *Gravitational Entropic Acceleration*. arXiv:2406.XXXXX [astro-ph.CO]. (See also: Calderon, R., et al. 2025.)
 - [9] Akrami, Y., et al. (COMPACT Collaboration) (2024). *Promise of Future Searches for Cosmic Topology*. Physical Review Letters 132, 171501. [doi:10.1103/PhysRevLett.132.171501](https://doi.org/10.1103/PhysRevLett.132.171501)
 - [10] Wurzel, S., & Hsu, S. (2025). *Progress toward fusion energy breakeven and gain as measured against the Lawson criterion*. Physics of Plasmas (updated). See also: Wikipedia, *Fusion energy gain factor* (accessed June 2026).
 - [11] ITER Physics Expert Groups et al. (1999). *ITER Physics Basis*. Nuclear Fusion 39, 2137. [doi:10.1088/0029-5515/39/12/308](https://doi.org/10.1088/0029-5515/39/12/308)
 - [12] Klinger, T., et al. (Wendelstein 7-X Team) (2025). *Overview of Wendelstein 7-X high-performance operation: OP 2.2 and OP 2.3 campaigns*. Presented at IAEA Fusion Energy Conference 2025. researchgate.net/publication/403811098
 - [13] Princeton Plasma Physics Laboratory (2025). *Wendelstein 7-X sets new performance records in fusion research*. PPPL News, June 2025. pppl.gov